12 **C** $W_{ext} = m\Delta\phi$

$50 = 2.0\left(\phi_B - \phi_A\right)$

$\phi_B - \phi_A = 25 \text{ J kg}^{-1}$

$W_{ext} = m\Delta\phi$

$-60 = 2.0\left(\phi_C - \phi_B\right)$

$\phi_C - \phi_B = -30 \text{ J kg}^{-1}$

$W_{ext} = m\Delta\phi$

$1000 = 2.0\left(\phi_\infty - \phi_C\right)$

$\phi_C = -500 \text{ J kg}^{-1}$

$\therefore \phi_B = -470 \text{ J kg}^{-1}$ and $\phi_A = -495 \text{ J kg}^{-1}$

13 **A** $Q = mc\Delta\theta$